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Government Fund Allocation and Web Tracking System

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Introduction

The Government Fund Allocation and Web Tracking System is a user-friendly platform that helps officials manage and track project funding, while allowing citizens to see how public money is spent. It promotes transparency, reduces fund misuse, and builds trust between the government and the public.

Aims

To build a simple and transparent web system that helps the government manage, track, and display fund allocation in real time, reduce misuse, and build public trust. Future integration of AI will help detect fraud and improve overall governance.

Objective

- To develop an interactive public dashboard for tracking fund distribution and project progress, enhancing transparency.
- Facilitate smooth communication and data exchange between government departments, officials, and citizens for improved project tracking and decision-making.

Academic Question

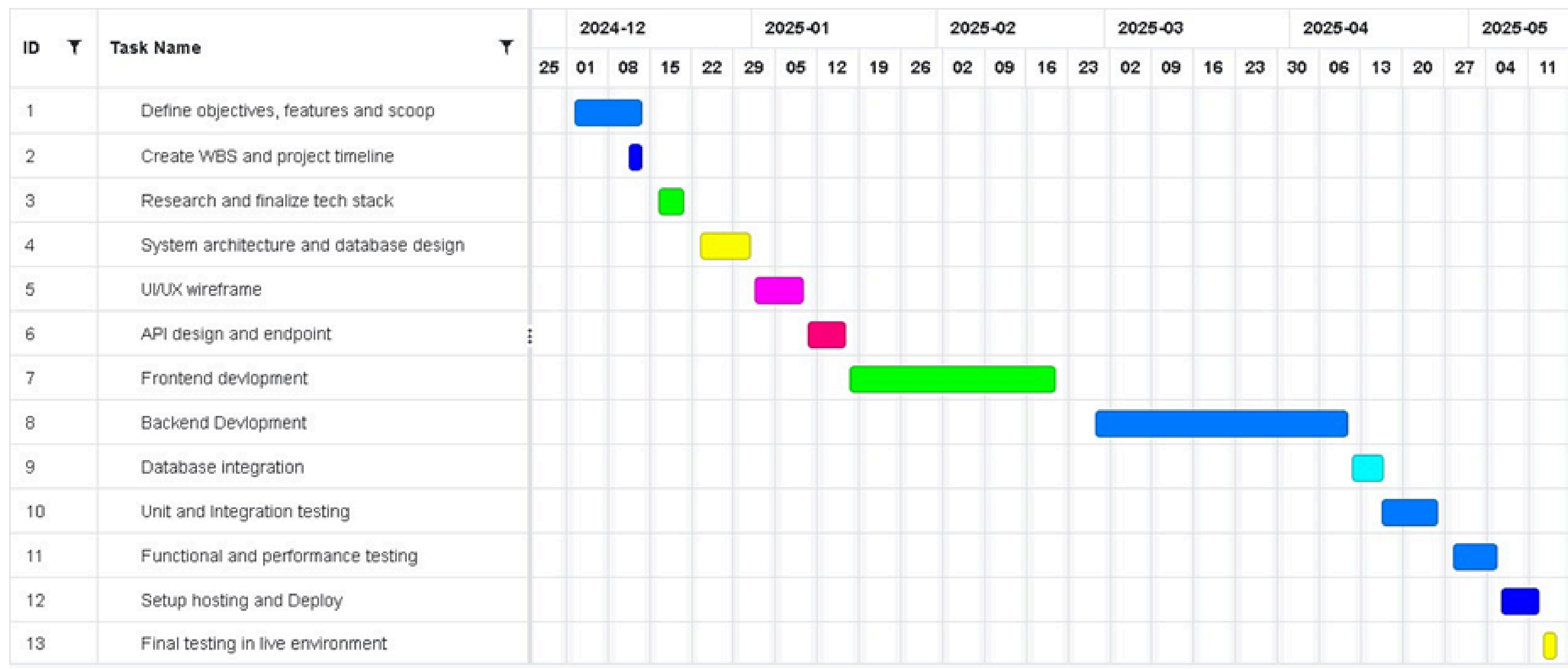
How does a Government Fund Allocation and Web Tracking System improve transparency, accountability and utilization of fund?

The system enhances transparency with real-time tracking, builds trust through notifications about delays and approvals, and ensures accountability It also integrates auditing systems to improve oversight and efficient fund management.

Literature Review

Title	Author (Year)	Key Features	Transparency	Fund Tracking	Fraud Detection
Allocation and Tracking Public Funds Using Blockchain	Kalpesh Bhole (2021)	Decentralized fund tracking, Smart contracts	High via immutable records	Automated via smart contracts	Reduced misuse risk
Optimization of Financial Shared Funds Management System Based on Cloud Computing	Xiufen Lu (2022)	Real-time financial tracking, Centralized database	Moderate via cloud visibility	Enhanced tracking of large funds	Not directly addressed
Application Algorithm for Cloud Environment Pooled Resource Scheduling and Allocation	Cai Di (2024)	Adaptive resource allocation, AI models	Moderate through efficiency	Real-time and predictive allocation	With predictive AI
Graph Feature Preprocessor for Financial Crime Detection	Jovan Blanuša (2024)	Real-time fraud detection using subgraphs	High for financial patterns	Based on transaction pattern	Advanced fraud detection
Financial Budget Model Based on Blockchain and Data Mining	Chunmiao Liu (2024)	Smart contracts + historical analysis	Very high with immutable logs	Based on real-time budget patterns	Detected through trends
Digital Government Ecosystem: Adaptive Architecture for ICT Investment	Asif Qumer Gill (2024)	Adaptive frameworks for digital governance	Through digital ecosystems	Through national-level planning	Not a core focus

Gantt Chart



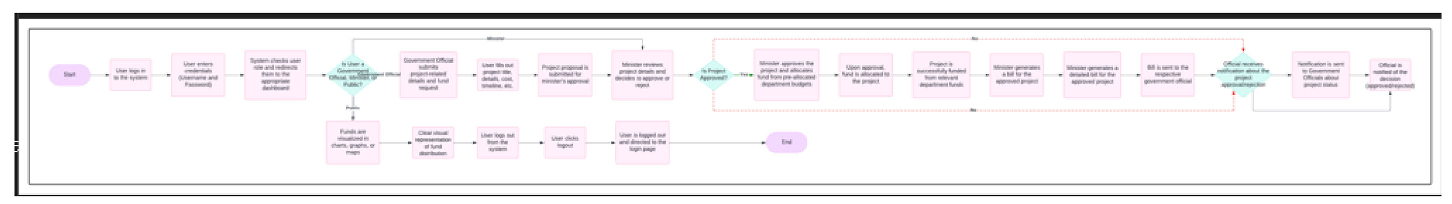
Future Scope

- Use AI to detect irregularities and potential fraud in real-time, improving governance and transparency.
- Integrate with existing government audit systems to streamline financial oversight and reporting.
- Implement blockchain to create immutable records of financial transactions, enhancing security and transparency.

Testing

Summary Point	Details
1. User Authentication & Access Control	All user roles (Ministers, Officials, Public) were successfully able to register and login securely. Role-based access control redirected users to their specific dashboards. Password strength and JWT authentication were verified. (TC-01 to TC-06)
2. Admin Functionalities	Admin was able to effectively manage user data including add, delete, and edit functions. (TC-03)
3. Fund Allocation Requests & Approvals	Government Officials successfully submitted detailed project fund requests. Ministers could approve or reject them. (TC-07, TC-08)
4. Fund Request Validation	Proper status tracking was implemented. Date validations and fund limits were also tested and functional, preventing overspending or past-date entries. (TC-09, TC-11, TC-12)
5. Departmental Fund Management	Fund requests were correctly allocated from respective departmental budgets with no cross-departmental conflicts. (TC-10)
6. Real-Time Tracking & Visualization	Real-time fund allocation and spending were displayed clearly using pie charts and graphs. Fund usage was dynamically updated. (TC-13 to TC-15)
7. Project Monitoring & History	Project history including allocation, requester, approver, and progress updates were properly recorded and visible. Progress tracking used a percentage-based question system. (TC-16 to TC-18)

Project Flow



Project Progress

Below I have listed the completed features of my project:

- Login
- Role-based access control
- Government official submits fund request for project
- Minister approves/rejects fund requests
- Fund allocation and tracking management
- Bill generation and sending to officials
- Notifications for approvals, rejections, and delays
- Project progress tracking
- Fund visualization (charts, graphs, maps)
- User management (role-based access)
- Data analysis and report generation

Findings and Conclusion

Findings:

1. The system offers real-time fund tracking and data visualization, allowing users to clearly see how public funds are allocated and spent.
2. Fund visualization tools (charts, graphs) make complex financial data easily understandable.
3. Government officials can submit projects with detailed information, and Ministers can approve or reject them efficiently, making fund distribution more systematic.

Conclusion: FundWatch enhances transparency and efficiency in public fund management through real-time tracking, role-based access, and data visualization, promoting better governance and public trust. Future improvements include AI fraud detection and audit system integration.